DIC08
Inverter DC Characteristics and Noise Margins

Specification-based laboratory report

17 October 2026

English edition
HSPICE S-2021.09 | Model library supplied separately

1 Objectives

Follow Lab 8 (PDF pages 31–34) to obtain inverter transfer curves, switching thresholds and noise margins, then test downstream regeneration using finite-duration disturbances.

2 Specification mapping

Handout requirement (PDF page)	Implementation	Verification and scope
Step 1: VTC, slope −1 and NM (31–33)	01_dc_transfer	Central derivative and interpolated crossings
Step 2: vary width ratio (32–33)	$W_p/W_n = 0.75, 3, 12$	All three curves
Step 3: noise at each ratio (33)	02_noise_cascade	Positive low-state and negative high-state disturbances
Question: add CMOS receiver (34)	Second stage for every ratio	Both stages retained

3 Circuit and design derivation

Fix $W_n = 0.6\,\mu$m and $L = 0.25\,\mu$m; $W_p = 0.45, 1.8, 7.2\,\mu$m implements the three ratios. The switching threshold is defined by $V_o = V_{DD}/2$, not an individual MOS threshold. V_{IL}, V_{IH} are the two $dV_o/dV_i = -1$ crossings. Boundary outputs $V_{OH} = V_o(V_{IL})$ and $V_{OL} = V_o(V_{IH})$ give $NM_L = V_{IL} - V_{OL}$ and $NM_H = V_{OH} - V_{IH}$. CSV files also retain the handout's ideal-rail NM approximation to keep definitions distinct. Partial derivative tests and scattered source sizes are replaced by complete two-boundary extraction and all three ratios.

4 Simulation setup

This teaching rewrite uses a separately supplied model at 1.0 V and 25 °C. The specified topology, loads and comparative dimensions are retained. The handout's 2.5 V process results are not accep-

tance values for this model. The chosen 1 V is a simulation condition, not a verified device rating. MOS bulk terminals are explicitly connected to the corresponding supply rails.

HSPICE S-2021.09 uses RUNLVL=6, retaining waveforms, printed data and native measurements. Figures are redrawn from actual data; CSV files are in each Lab's results directory. Successful solution does not establish process reliability. Warnings from encrypted model blocks remain suppressed; the existing diagnostic did not expose actionable further details, so the warnings are not declared harmless and complete model-limit compliance is not claimed.

DC sweeps 0–1 V in 0.5 mV steps. The 18 ns transient prints every 2 ps. Low-state disturbances reach 0.2/0.4/0.48/0.6 V; high-state dips reach 0.8/0.6/0.52/0.4 V. Edges are 0.1 ns and plateaus 0.9 ns. This normalized test preserves the noise experiment without applying the original 1.2/1.3 V thresholds to a 1 V supply.

5 Results and specification checks

Both decks completed. Voltages and margins below are in V and use boundary output levels.

W_p/W_n	V_M	V_{IL}	V_{IH}	NM_L	NM_H
0.75	0.4214	0.3617	0.4496	0.3237	0.5100
3	0.4825	0.4291	0.5189	0.3855	0.4376
12	0.5459	0.5031	0.5925	0.4600	0.3616

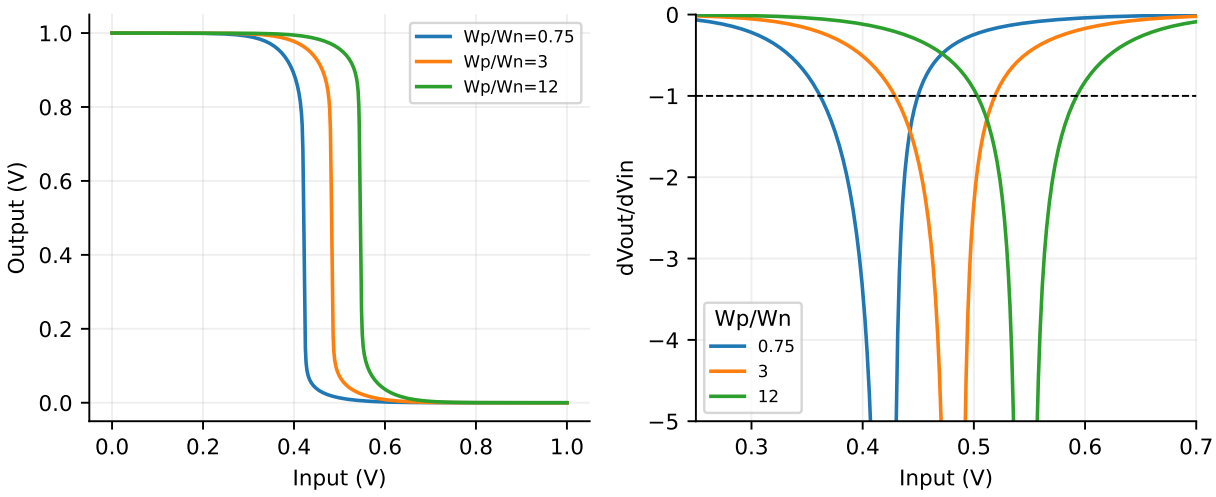


Figure 1: Transfer curves and unity-gain boundaries; redrawn from this run's HSPICE data.

6 Discussion

Widening PMOS shifts the threshold upward, increasing NM_L and decreasing NM_H; ratio 3 is not exactly symmetric in this model. A second stage can restore disturbances that stay outside its switching region, but may regenerate a wrong logic excursion into a full pulse. Filtering depends on duration and loading; transient survival is not identical to DC noise margin.

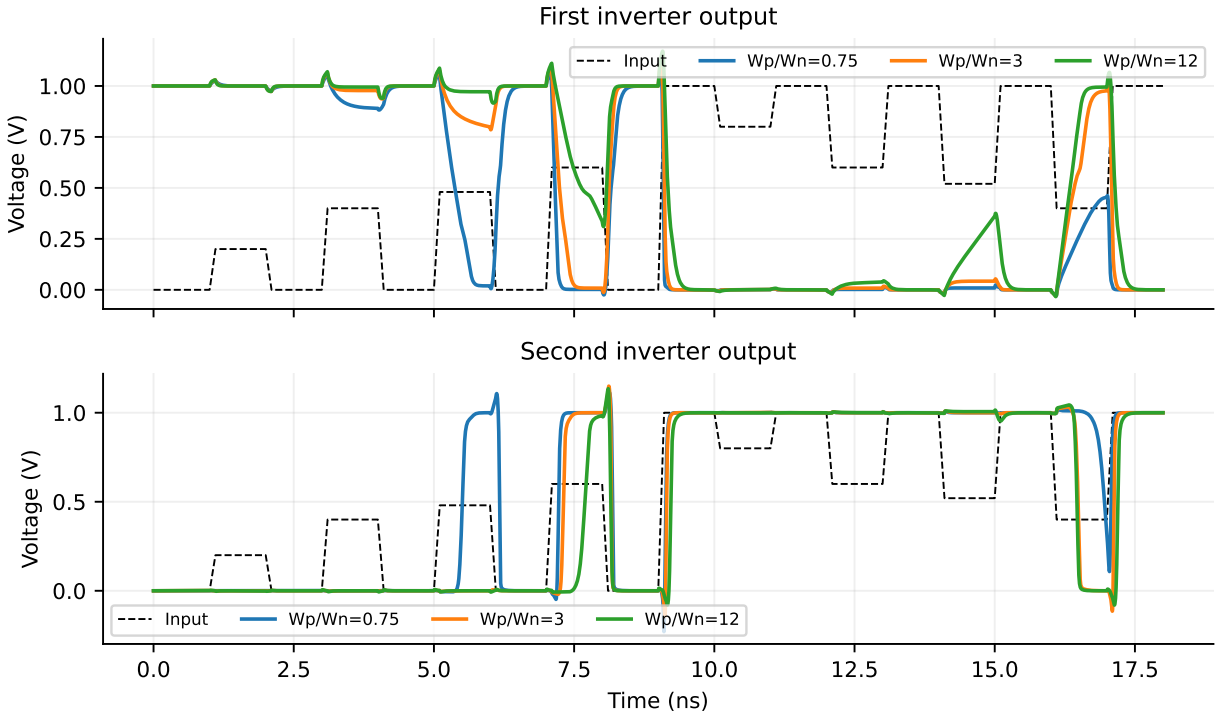


Figure 2: Noise responses at the first and second stages for all ratios; redrawn from this run's HSPICE data.

7 Conclusion

All three steps and the cascade question have complete simulation evidence. Voltage normalization and the NM definition are explicit; original-process numerical margins are not claimed.